

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 2– Code Debugging Questions**

|                  |                                                                                                                                                               |               |                                                          |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------|
| Course Code:     | CSA0303                                                                                                                                                       | Course Title: | Data Structures                                          |
| CO Assessed:     | CO2                                                                                                                                                           | Assessment:   | Assessment Tool 2 – Debugging and Optimization Questions |
| Weightage:       | 10% of CIA                                                                                                                                                    | Total Marks:  | 100                                                      |
| CO2 Statement:   | ADT array, Operations on arrays, Searching Algorithms, Linear Search, Binary Search, Suffix Arrays & LCP Arrays ADT, Linked List, Stack, Queue, Applications. |               |                                                          |
| Student Name:    | ALURI POOJITHA                                                                                                                                                | Reg. No.:     | 192511230                                                |
| Section / Batch: | CSE                                                                                                                                                           | Date:         |                                                          |

**Code of Conduct**ALURI POOJITHA

I \_\_\_\_\_ALURI POOJITHA\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_POOJITHA\_\_\_\_\_

**Instructions**

- This test contains 50 MCQ Based Questions, each carries 2 marks. Total: 100 Marks.
- When a student answer using a analytical problem, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are not permitted in the class. Take it as test and submit in the class.

| Section / Topic | Focus                       | Q Nos     | Marks |
|-----------------|-----------------------------|-----------|-------|
| Stack           | Linear                      | 10        | 30    |
| Queue           | Enq, Deque                  | 10        | 40    |
| Linked List     | Applications of linked list | 10        | 40    |
| GRAND TOTAL:    |                             | 100 Marks |       |

**Annexure A – Code Debugging Questions**

**\*\*>> Enclosed**

## Rubrics for Calculation

| Criteria                         | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |
| Faculty In-charge                |           | Course Coordinator                                                         |                                                       | Program Director                               |                                                        |
| Signature: _____                 |           | Signature: _____                                                           |                                                       | Signature: _____                               |                                                        |
| Date: _____                      |           | Date: _____                                                                |                                                       | Date: _____                                    |                                                        |

## CO2-AT2-Code Debugging

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### QUESTIONS

**Q1** Find the bugs in the following snippet.  
2 marks

**Q2** Identify the error in the following snippet.  
2 marks

**Q3** Find the bugs in the given snippet.  
2 marks

**Q4** Identify the error in the following snippet.  
2 marks

**Q5** Where is the error? Identify it.  
2 marks

**Q30** Identify the truth in time complexity analysis.

2 marks

Identify the truth in time complexity analysis. Identify the truth in time complexity analysis.

A. Linear search is always faster

B. Binary search is  $O(\log n)$  only for sorted data

C. Stack operations are  $O(n)$

D. Queue operations are  $O(\log n)$

Save Answer

Next →

## 2-AT2-Code Debugging

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### QUESTIONS

**Q1** Find the bugs in the following snippet.  
2 marks

**Q2** Identify the error in the following snippet.  
2 marks

**Q3** Find the bugs in the given snippet.  
2 marks

**Q4** Identify the error in the following snippet.  
2 marks

**Q5** Where is the error? Identify it.  
2 marks

**Q1** Find the bugs in the following snippet.

2 marks

Find the bugs in the following snippet. `int linearSearch(int arr[], int n, int key) {for(int i = 0; i <= n; i++) {if(arr[i] == key)return i;}return -1;}`

A. Wrong return type

B. Loop condition is incorrect

C. Missing break statement

D. Array not initialized

Save Answer

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## CO2-AT2-Code Debugging

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### QUESTIONS

**Q1** Find the bugs in the following snippet.  
2 marks

**Q2** Identify the error in the following snippet.  
2 marks

**Q3** Find the bugs in the given snippet.  
2 marks

**Q4** Identify the error in the following snippet.  
2 marks

**Q5** Where is the error? Identify it.

**Q6** Identify the error and give me the correct answers.

2 marks

Identify the error and give me the correct answers. `#define MAX 5int stack[MAX];int top = -1;void push(int x) {if(top == MAX) {printf("Overflow");return;}stack[++top] = x;}`

A. Stack size is wrong

B. Overflow condition is incorrect

C. top initialization wrong

D. push syntax incorrect

Save Answer

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## CO2-AT2-Code Debugging

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### QUESTIONS

**Q1** Find the bugs in the following snippet.  
2 marks

**Q2** Identify the error in the following snippet.  
2 marks

**Q3** Find the bugs in the given snippet.  
2 marks

**Q4** Identify the error in the following snippet.  
2 marks

**Q5** Where is the error? Identify it.

**Q8** What is the issue in the following snippet?

2 marks

What is the issue in the following snippet? `void push(int x) { stack[top++] = x;}`

A. Syntax error

B. Wrong increment order

C. Missing return

D. No overflow check

Save Answer

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## CO2-AT2-Code Debugging

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## QUESTIONS

Q1

Find the bugs in the following snippet.  
2 marks



Q2

Identify the error in the following snippet.  
2 marks



Q3

Find the bugs in the given snippet.  
2 marks



Q4

Identify the error in the following snippet.  
2 marks



Q5

Where is the error? Identify it.  
2 marks



Q6



Q23 What is the issue?

2 marks

```
What is the issue? temp = head
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

A. Infinite loop

B. Last node not printed

C. First node skipped

D. Syntax error

Save Answer

Next →